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CURRENT  
POSITION

2025-      University of California, Irvine  
              Associate Professor, Department of Cognitive Sciences  
              Department of Neurobiology and Behavior (by courtesy)  
 Center affiliations:  
              Center for the Neurobiology of Learning and Memory  
              Center for Theoretical Behavioral Sciences  
              Drury Institute for Translational Depression Discoveries  
              Center for Early Life Environment and Neuropsychiatric Outcomes

PREVIOUS

APPOINT-  
MENTS

2019-2025 Assistant Professor, [Department of Cognitive Sciences](#), UC Irvine  
 2018       Associate Research Scholar, Princeton Neuroscience Institute  
 2013-2018 Postdoctoral Research Associate, Princeton Neuroscience Institute  
 2007-2013 Graduate researcher, New York University  
 2006       Research assistant, Deicken lab, UCSF/VA  
 2005-2007 Research assistant, Wagner lab, Stanford

EDUCATION

2013    Ph.D., Cognition & Perception, **New York University**  
 2003    S.B., Mathematics (additional concentration in Economics)  
          **Massachusetts Institute of Technology**

RESEARCH  
FUNDING

*Ongoing*

2026-2027    UCLA-CDU Dana Center Seed Grant (MPI with Carr, Jones, & Kalofonos)  
 “Critically examining neuroscience evidence for anosognosia as a transdiagnostic symptom”  
 2025-2030    NIMH R01MH061285 (subaward; PI SD Pollak)  
 “Emotion Processing: Risk for Psychopathology in Children”  
 2025-2030    NIA R01AG088306 (PI; MPI IJ Bennett)  
 “Memory-guided decision making across the lifespan”  
 2023-2028    NIMH R01MH128306 (Co-I; PI MA Yassa)  
 “Testing a memory-based hypothesis for anhedonia”  
 2022-2027    NINDS R01NS119468 (Co-I; PI ER Chrestil)  
 “Cognitive graphs, active decision making, and brain network dynamics”

### *Completed*

- 2022-2025 NIMH P50MH096889 (Co-I; PI TZ Baram)  
“Fragmented early-life experiences, aberrant circuit maturation, emotional vulnerabilities”
- 2022-2025 UK MRC MR/W028476/1 (Collaborator; PI M Field)  
“Reinforcer-specific value-based decision-making in persistence of and recovery from alcohol use disorder”
- 2021-2025 NIA R21AG072673 (PI)  
“Improving multi-step planning in aging by overcoming deficits in memory encoding”
- 2021-2023 BBRF NARSAD Young Investigator award (PI)  
“Determining the role of episodic memory in substance use disorder”
- 2020-2021 NIMH P50MH096889 (seed grant; PI TZ Baram)  
“Early life adversity effects on event segmentation”
- 2011-2013 NIMH F31MH095501 (PI)  
“Computational mechanisms of goal-directed control”

### *To lab members/trainees*

- 2027-2029 BBRF NARSAD Young Investigator award (Dale Zhou)
- 2026 UCI FIBRE (Neuroimaging center) project award (Rohin Palsule)
- 2026-2028 NIMH T32MH119049 (Alexa Booras)
- 2026 UROP Undergraduate Research Support Award (Dhriti Grover)
- 2026 UROP Undergraduate Research Support Award (Aidan Goeschel & Bryce Wilson)
- 2026 UROP Undergraduate Research Support Award (Annie Shen)
- 2025 NIA/SRNDNA Collaboration award (Shensheng Wang)
- 2025 UROP Undergraduate Research Support Award (Jianle Guo & Rohin Pasule)
- 2024 ARCS Foundation Scholar (Bianca Leonard)
- 2024 UCI Summer Undergraduate Research Fellowship (Meghan Johnson)
- 2024 Indow Fellowship for Research Excellence (Jungsun Yoo)
- 2024 NIA/SRNDNA Summer Research award (Melisa Azimihashemi)
- 2024 NIA/SRNDNA Summer Research award (Ami Yamamoto)
- 2024 UROP Research Experience Fellowship (Ami Yamamoto)
- 2023-2026 Hewitt Foundation postdoctoral fellowship (Dale Zhou)
- 2023-2025 NIMH F31MH134620 (Nora Harhen)
- 2023-2025 NIMH T32MH119049 (Ari Khoudary)
- 2023 Robert J. Glushko prize for best undergraduate thesis (Gloria Cheng)
- 2023 CNLM Jared M. Roberts Graduate Student Award (Ari Khoudary)
- 2023 CNLM Jared M. Roberts Graduate Student Award (Nidhi Banavar)
- 2022 UCI Summer Undergraduate Research Fellowship (Gloria Cheng)
- 2022 Lambert prize in Foundations of Science (Nidhi Banavar)
- 2022-2025 NIA F32AG072836 (Sharon Noh)
- 2020-2023 National Defense Science & Engineering Graduate fellowship (Nora Harhen)
- 2019 UCI Summer Undergraduate Research Fellowship (Brianna Sarcos)

## PUBLICATIONS

(\* = *Equal contribution*; + = *UCI Lab member*.)

### *Preprints*

- PP5 Leonard BT+, Martínez-Ortíz MA, Bock J, Zhang Y+, Flores S, Taylor DV, Glynn LM, Davis EP, Stern HS, Baram TZ, Hartley CA, Yassa MA\*, **Bornstein AM\***. Anhedonia buffers the effects of early-life unpredictability on threat-reward decision-making. *bioRxiv*.  
doi:10.64898/2026.05.16.725643
- PP4 Wang W, Chierchia G, Cooley BJ, Chang T, **Bornstein AM**, Schweizer S. Autobiographical Memory Specificity Shapes Navigation of Social Uncertainty in Depression. *PsyArXiv*.  
doi:10.31234/osf.io/z6vej
- PP3 Yoo J+, **Bornstein AM**. Temporal dynamics of model-based control reveal arbitration between multiple task representations. *PsyArXiv*.  
doi:10.31234/osf.io/sgcy5
- PP2 Zhang Y+\*, Chen Y+\*, Chen XR+, Harhen NC+, Glynn LA, Davis EP, Baram TZ, Risbrough VB, Stout DA, **Bornstein AM**. A specific computational role for early-life, and not lifelong, stress in decision-making under uncertainty. *bioRxiv*.  
doi:10.64898/2026.06.01.729407
- PP1 Zhou D+, Noh SM+, Harhen NC+, Banavar NV+, Kirwan B, Yassa MA, **Bornstein AM**. A compressed code for memory discrimination. *bioRxiv*.  
doi:10.1101/2025.10.12.681901

### *Peer-reviewed journal articles*

- JP28 Harhen NC+, **Bornstein AM\***, Hartley CA\* (2026). Developmental changes in memory structure and precision alter the use of retrieved episodes during decisions for reward. *Proceedings of the National Academy of Sciences*, 123(31):e2525494123.  
doi:10.1073/pnas.2525494123
- JP27 Harhen NC+, Budiono R, Hartley CA\*, **Bornstein AM\*** (2026). Structure inference in complex environments improves from childhood to adulthood. *Developmental Science*, 29(3):e70163.  
doi:10.1111/desc.70163
- JP26 Noh SM+\*, Zhou D+\*, Cooper KW, Guo S+, Dinh ET+, **Bornstein AM** (2026). Sparsity and memory constraints interact with training sequence to bias learning of associative maps. *Neuropsychologia*, 225:109396.  
doi:10.1016/j.neuropsychologia.2026.109396
- JP25 Noh SM+\*, Cooper KW\*, Guo S+, Zhou D+, Stark CEL, **Bornstein AM** (2026). Multi-step inference can be improved across the human lifespan with individualized memory interventions. *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, 81(5):gbag038.  
doi:10.1093/geronb/gbag038
- JP24 Hadj-Amar B, **Bornstein AM**, Guindani M, Vannuci M (2025). Discrete Autoregressive Switching Processes with Cumulative Shrinkage Priors for Graphical Modeling of Time Series Data. *Journal of Computational and Graphical Statistics*.  
doi:10.1080/10618600.2025.2551271

- JP23 Schetzle B, Lee J, **Bornstein AM**, Shahbaba B, Guindani M (2025). A Bayesian Time-Varying Psychophysiological Interaction Model. *Data Science in Science*, 4(1):2519436. doi:10.1080/26941899.2025.2519436
- JP22 Khoudary A+, Peters MAK\*, **Bornstein AM\*** (2025). Reasoning goals and representational decisions in computational cognitive neuroscience: lessons from the drift diffusion model. *European Journal of Neuroscience*, 61:e7009. doi:10.1111/ejn.70098
- JP21 Banavar NV+, Noh SM+, Wahlheim CN, Cassidy BS, Kirwan CB, Stark CEL, **Bornstein AM** (2024). A response time model of the three-choice Mnemonic Similarity Task provides stable, mechanistically interpretable individual-difference measures. *Frontiers in Human Neuroscience*, 18:137928. doi:10.3389/fnhum.2024.137928
- JP20 Yoo J+, Chrastil ER, **Bornstein AM** (2024). Cognitive graphs: Representational substrates for planning. *Decision*, 11(4), 537-556. doi:10.1037/dec0000249
- JP19 Chen J, **Bornstein AM** (2024). The causal structure and computational value of narratives. *Trends in Cognitive Sciences*, 28(8):769-781. doi:10.1016/j.tics.2024.04.003
- JP18 Harhen NC+, **Bornstein AM** (2024). Interval timing as a computational pathway from early life adversity to affective disorders. *Topics in Cognitive Science*, 16(2024):92-112. doi:10.1111/tops.12701
- JP17 Noh SM+, Singla UK, Bennett IJ, **Bornstein AM** (2023). Memory precision and age differentially predict the use of decision-making strategies across the lifespan. *Scientific Reports*, 13:17014. doi:10.1038/s41598-023-44107-5
- JP16 **Bornstein AM**, Aly M, Feng SF, Turk-Browne NB, Norman KA, Cohen JD (2023). Associative memory retrieval modulates upcoming perceptual decisions. *Cognitive, Affective, & Behavioral Neuroscience*, 23:645665. doi:10.3758/s13415-023-01092-6 doi:10.18112/openneuro.ds001614.v1.0.1
- JP15 Harhen NC+, **Bornstein AM** (2023). Overharvesting in human patch foraging reflects rational structure learning and adaptive planning. *Proceedings of the National Academy of Sciences*, 120(13):e2216524120. doi:10.1073/pnas.2216524120
- JP14 Otto AR, Devine S, Schultz E, **Bornstein AM\***, Louie K\* (2022). Context-dependent choice and evaluation in real-world consumer behavior. *Scientific Reports*, 12:17744. doi:10.1038/s41598-022-22416-5 doi:10.17605/OSF.IO/EC5DX
- JP13 Rmus M, Ritz H, Hunter LE, **Bornstein AM\***, Shenhav A\* (2022). Humans can navigate complex graph structures acquired during latent learning. *Cognition*, 225:105103. doi:10.1016/j.cognition.2022.105103
- JP12 Wang S, Feng SF, **Bornstein AM** (2021). Mixing memory and desire: How decisions for reward depend on the dynamics and content of memory reinstatement. *Wiley Interdisciplinary Reviews: Cognitive Science*, e1581. doi:10.1002/wcs.1581

**Featured on the cover.**

**Recognized as one of the top-ten most-cited papers from this journal/year.**

- JP11 Rouhani N, Norman KA, Niv Y, **Bornstein AM** (2020). Reward prediction errors create event boundaries in memory. *Cognition*, 203:104269.  
doi:10.1016/j.cognition.2020.104269
- JP10 **Bornstein AM\***, Pickard H\* (2020). Chasing the first high: Memory sampling in drug choice. *Neuropsychopharmacology*, 45(6):907-915.  
doi:10.1038/s41386-019-0594-2 **Featured on the cover.**
- JP9 Kane GA, **Bornstein AM**, Shenhav A, Wilson RC, Daw ND, Cohen JD (2019). Rats exhibit similar biases in foraging and intertemporal choice tasks. *eLife*, 8:e48429.  
doi:10.7554/eLife.48429
- JP8 Millner AJ, den Ouden HEM, Gershman SJ, Glenn CR, Kearns J, **Bornstein AM**, Marx BP, Keane TM, Nock MK (2019). Suicidal thoughts and behaviors are associated with an increased decision-making bias for active responses to escape aversive states. *Journal of Abnormal Psychology*, 128(2):106-118.  
doi:10.1037/abn0000395
- JP7 Hoskin AN, **Bornstein AM**, Norman KA, Cohen JD (2018). Refresh my memory: Episodic memory reinstatements intrude on working memory maintenance. *Cognitive, Affective, & Behavioral Neuroscience*, 19:338-354.  
doi:10.3758/s13415-018-00674-z  
doi:10.18112/openneuro.ds001576.v1.0.0
- JP6 **Bornstein AM**, Khaw MW, Shohamy D, Daw ND (2017). Reminders of past choices bias decisions for reward in humans. *Nature Communications*, 8:15958.  
doi:10.1038/ncomms15958
- JP5 **Bornstein AM**, Norman KA (2017). Reinstated episodic context guides sampling-based decisions for reward. *Nature Neuroscience*, 20:997-1003.  
doi:10.1038/nn.4573  
doi:10.18112/openneuro.ds001607.v1.0.1
- JP4 **Bornstein AM**, Daw ND (2013). Cortical and hippocampal correlates of deliberation during model-based decisions for rewards in humans. *PLoS Computational Biology*, 9(12):e1003387.  
doi:10.1371/journal.pcbi.1003387
- JP3 **Bornstein AM**, Daw ND (2012). Dissociating hippocampal and striatal contributions to sequential prediction learning. *European Journal of Neuroscience*, 35:1011-1023.  
doi:10.1111/j.1460-9568.2011.07920.x
- JP2 **Bornstein AM**, Daw ND (2011). Multiplicity of control in the basal ganglia: computational roles of striatal subregions. *Current Opinion in Neurobiology*, 21(3):374-380.  
doi:10.1016/j.conb.2011.02.009
- JP1 Preston AR, **Bornstein AM**, Hutchinson JB, Gaare ME, Glover GH, Wagner AD (2010). High-resolution fMRI of content-sensitive subsequent memory responses in human medial temporal lobe. *Journal of Cognitive Neuroscience*, 22:156-173.  
doi:10.1162/jocn.2009.21195

*Articles in conference proceedings*

- CP17 Zhou D, Irizarry-Márquez G, Guo S, Kim S, Janecek J, Tuteja N, Meza N, McMillan L, Thayer J, **Bornstein AM\***, Yassa MA\* (2026). Effects of surprising rewards on pattern separation. *Proceedings of the 48th Annual Conference of the Cognitive Science Society*.
- CP16 Khoudary A+, **Bornstein AM\***, Peters MAK\* (2025). Subjective and objective cue probability interact to shape perceptual decisions. *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.
- CP15 Chen Y+, Harhen NC+, Stout DA, **Bornstein AM** (2024). Early life unpredictability modulates planning horizon in a structured foraging task. *Cognitive Computational Neuroscience*.
- CP14 Harhen NC+, Hartley CA\*, **Bornstein AM\*** (2024). Development of structure inference contributes to age-related differences in exploration. *Cognitive Computational Neuroscience*.
- CP13 Yoo J+, Zhou D+, **Bornstein AM** (2024). Latent cause inference as an efficient and flexible learning rule for cognitive graphs. *Cognitive Computational Neuroscience*.
- CP12 Zhou D+, Noh SM+, Yassa MA, **Bornstein AM** (2024). Pattern separation using compressed and semantic representations of memory. *Cognitive Computational Neuroscience*.
- CP11 Khoudary A+, Peters MAK\*, **Bornstein AM\*** (2022). Precision-weighted evidence integration predicts time-varying influence of memory on perceptual decisions. *Cognitive Computational Neuroscience*.
- CP10 Banavar NV+, **Bornstein AM** (2022). Response time modeling provides stable and mechanistically interpretable measures of individual differences in behavioral pattern separation. *Proceedings of the 20th International Conference on Cognitive Modeling*.  
**Selected for a talk.**
- CP9 Harhen NC+, **Bornstein AM** (2022). Learning to expect change: Volatility during early experience alters reward expectations in a model of interval timing. *Proceedings of the 20th International Conference on Cognitive Modeling*.  
**Selected for a talk.**  
**Selected for a best paper award.**
- CP8 Banavar NV+, **Bornstein AM** (2022). Decision difficulty modulates the re-use of computations across trials in non-sequential decision tasks. *Proceedings of the 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM 2022)*
- CP7 Harhen NC+, **Bornstein AM** (2022). Humans adapt their foraging strategies and computations to environment complexity. *Proceedings of the 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM 2022)*
- CP6 Yoo J+, **Bornstein AM** (2022). Two-stage task with increased state space complexity to assess online planning. *Proceedings of the 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM 2022)*
- CP5 Banavar NV+, Lee MD, **Bornstein AM** (2021). Sequential effects in non-sequential tasks. *Proceedings of the 19th International Conference on Cognitive Modeling*.
- CP4 Harhen NC+, **Bornstein AM** (2021). Structure learning as a mechanism of overharvesting. *Proceedings of the 19th International Conference on Cognitive Modeling*.
- CP3 Harhen NC+, Hartley CA, **Bornstein AM** (2021). Model-based foraging using latent-cause inference. *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*.

doi:10.31234/osf.io/dfztu

<https://github.com/uciccnl/CogSci2021-HarhenHartleyBornstein>

- CP2 Kane GA, **Bornstein AM**, Shenhav A, Wilson RC, Daw ND, Cohen JD (2017). Mechanisms of overharvesting in patch foraging. *Proceedings of the 39th Annual Conference of the Cognitive Science Society*, 637-642.
- CP1 Floares A, Jakary A, **Bornstein A**, Deicken R (2006). Neural networks and classification and regression trees are able to distinguish females with major depression from healthy controls using neuroimaging data. *Proceedings of the IEEE International Joint Conference of Neural Networks, 2006*, 4605-4611.  
doi:10.1109/ijcnn.2006.247090

### *Commentaries & Book Chapters*

- RP8 Zhou D+, **Bornstein AM** (in press). Decompressing the kinds of compression (Commentary on Dubova & Sloman). *Behavioral and Brain Sciences*.  
doi:10.1017/S0140525X26107511
- RP7 Khoudary A+, **Bornstein AM\***, Peters MAK\* (2026). Blind ignorance or rational inattention? On the history and future of metabolic considerations for cognitive models (Commentary on Haueis & Colaco). *Behavioral and Brain Sciences*.  
doi:10.1017/S0140525X26106293
- RP6 Zhou D+, **Bornstein AM** (2024). Expanding horizons in reinforcement learning for curious exploration and creative planning (Commentary on Ivancovsky et al). *Behavioral and Brain Sciences*, 47, e118.  
doi:10.1017/S0140525X23003394
- RP5 Banavar NV+, **Bornstein AM** (2024). Multi-plasticities: Distinguishing context-specific habits from complex perseverations. In Y. Vandaele (Ed.). *Habits - Their Definition, Neurobiology and Role in Addiction*. Cham, Switzerland: Springer Nature.
- RP4 **Bornstein AM\***, Constantino SM\* (2017). Nudge back: Towards a taxonomy of scientific rationalities. *London Conference in Critical Thought*.
- RP3 **Bornstein AM**, Miller KJ, Shenhav A (2015). Walking bundles of habits (and Response-Outcome associations). *European Journal of Neuroscience*, 41:1356-1357.  
doi:10.1111/ejn.12906
- RP2 **Bornstein AM** (2014). Functions of the hippocampal memory system in instrumental control (Doctoral dissertation).  
Available from ProQuest Dissertations & Theses Global (3614853).
- RP1 **Bornstein AM**, Nylen EL, Steele SA (2011). Unblocking the neural substrates of model-based value. *Journal of Neuroscience*, 31(28):10117-10118.  
doi:10.1523/jneurosci.1883-11.2011

### *Accepted conference abstracts (since 2019)*

- CA69 Noh SM+, Morton NW, **Bornstein AM**, Preston AR (2026). Interleaving supports category generalization by integrating exemplars and differentiating categories. *Psychonomics*.  
**Selected for a talk.**
- CA68 Grover D+, Leonard BT+, **Bornstein AM\*** (2026). Reaction Time and Behavioral Avoidance in Anxiety: Evidence from Human Prey Foraging. *Society for Neuroeconomics Annual Meeting*.

- CA67 Shen A+, **Bornstein AM\*** (2026). Joint Modeling of Response Time and Choice Reveals Asymmetric Switch Costs Under Risk and Ambiguity. *Society for Neuroeconomics Annual Meeting*.
- CA66 Zhou D+, Guo S+, Yassa MA, **Bornstein AM\*** (2026). Representation learning and choice in naturalistically complex environments. *Society for Neuroeconomics Annual Meeting*.
- CA65 Zhou D+, Harhen NC+, Davis EP, Glynn LM, Baram TZ, Risbrough VB, Stout DM, Yassa MA\*, **Bornstein AM\*** (2026). Early-Life Unpredictability and the Neural and Computational Timescales of Integration. *The Flux Society Congress for Integrative Developmental Neuroscience*.  
**Selected for a talk.**
- CA64 Booras AK, Kapogianis TK, Yoo J, **Bornstein AM\***, Chrastil ER\* (2026). Individual Variation in Non-Spatial Navigation is Related to Adaptive Learning and Use of Successor Representations. *Cognitive Computational Neuroscience*.
- CA63 Goeschel A\*, Palsule R\*, Yoo J, Chen J, **Bornstein AM**, Khoudary A (2026). Quantifying information gain in narrative stimuli using open-source large language models. *Cognitive Computational Neuroscience*.
- CA62 Guo S\*, Booras AK\*, Kahana MJ, Fried I, Sakon J, **Bornstein AM** (2026). Replay Supports Both Future and Immediate Decisions by Updating a Successor Representation. *Cognitive Computational Neuroscience*.
- CA61 Chen XR\*, Johnson MJ\*, Palsule RA, Maxwell M, Harhen NC, Noh SM, Bennett IJ, **Bornstein AM** (2026). Age-related differences in structure learning drive foraging behavior in a multimodal patch-leaving task. *Cognitive Computational Neuroscience*.
- CA60 Zhou D, Guo S, Yassa MA, **Bornstein AM** (2026). Representation learning and choice in naturalistically complex environments. *Cognitive Computational Neuroscience*.
- CA59 Zhang Y, **Bornstein AM** (2026). An adaptive role for anhedonia following early-life unpredictability. *Cognitive Computational Neuroscience*.
- CA58 Zhang Y, **Bornstein AM** (2026). Belief-space warping as an adaptive response to early-life unpredictability. *Computational Psychiatry 2026*.
- CA57 Zhou D+, Harhen NC+, Stout DM, Jirsaraie R, Parker D, Risbrough V, Davis EP, Glynn LM, Baram TZ, Yassa MA\*, **Bornstein AM\*** (2026). Adaptation to early life unpredictability: elongated temporal representations for reward-based decision making. *Computational Psychiatry 2026*.  
**Selected for a short talk.**
- CA56 Booras AK+, **Bornstein AM\***, Chrastil ER\* (2026). Non-spatial navigation is driven by both adaptive learning and use of successor representations. *CNLM Annual Meeting*.  
**Selected for a short talk.**
- CA55 Guo S+, Booras AK+, Sakon J, **Bornstein AM** (2026). Replay supports future decisions by updating a successor representation. *CNLM Annual Meeting*.  
**Selected for a short talk.**
- CA54 Palsule R+, Goeschel A+, Yoo J+, Chen J, **Bornstein AM** (2026). Comparing a large language model-derived metric of contextual information gain with human judgements in narratives. *CNLM Annual Meeting*.
- CA53 Zhou D+, Harhen NC+, Davis EP, Glynn LM, Baram TZ, Risbrough VB, Stout DM, Yassa MA\*, **Bornstein AM\*** (2026). Early-Life Unpredictability and the Neural and

Computational Timescales of Integration. *Anxiety and Depression Association Annual Meeting*.

- CA52 Leonard BT+, Martínez-Ortiz MA, Hartley CA, Yassa MA\*, **Bornstein AM\*** (2026). Anhedonia and early life unpredictability distinctly shape motivational conflict resolution in adults. *American College of Neuropsychopharmacology*.
- CA51 Booras A+, Chrastil ER\*, **Bornstein AM\*** (2025). Humans adaptively discount successor representations when navigating graph-structured spaces. *Bay Area Memory Meeting*.
- CA50 Khoudary A+, Peters MAK\*, **Bornstein AM\*** (2025). Uncertainty resolution explains dynamic effects of memory on perceptual decisions *Bay Area Memory Meeting*.  
**Selected for a talk.**
- CA49 Zhou D+, Yassa MA\*, **Bornstein AM\*** (2025). A lossy compression account of mnemonic discrimination: Computational theory and multimodal evidence. *Bay Area Memory Meeting*.  
**Selected for a talk.**
- CA48 Booras A+, Chrastil ER, **Bornstein AM** (2025). Humans adaptively discount successor representations when navigating graph-structured spaces. *Society for Neuroscience Annual Meeting*.
- CA47 Guo S+\*, Palsule R+\*, Noh SM+\*, Preston AR, **Bornstein AM** (2025). The impact of learning sequences in encouraging distinct memory representations to support associative inference in older-adults. *Society for Neuroscience Annual Meeting*.
- CA46 Khoudary A+, **Bornstein AM\***, Peters MAK\* (2025). A philosophical toolkit for computational modeling of cognition. *Society for Neuroscience Annual Meeting*.
- CA45 Khoudary A+, Peters MAK\*, **Bornstein AM\*** (2025). Precision-weighted integration explains dynamic effects of expectations on perceptual decisions. *Society for Neuroscience Annual Meeting*.
- CA44 Leonard BT+, Martínez-Ortiz MA, Hartley CA, Yassa MA, **Bornstein AM** (2025). Motivational conflict resolution is shaped by early life unpredictability. *Society for Neuroscience Annual Meeting*.
- CA43 Palsule R+\*, Guo S+\*, Noh SM+\*, **Bornstein AM** (2025). Naturalistic task framing improves older adults' ability to infer and navigate complex associative networks. *Society for Neuroscience Annual Meeting*.
- CA42 Yoo J+, Zhou D+, Goeschel A+, **Bornstein AM** (2025). Contingency-dependent state augmentation as a normative learning rule for non-Markovian tasks. *Society for Neuroscience Annual Meeting*.
- CA41 Field M, Copeland A, Hogarth L, Pun M, Ryan F, Pickard H, **Bornstein AM**, Murphy J, Strickland J (2025). Meaning in life and value-based decision-making in alcohol use disorder. *55th EABCT Congress*.
- CA40 Zhou D\*, Noh SM\*, Cooper KW, Guo S, Dinh ET, **Bornstein AM** (2025). Sparse and distributed memory constraints drive representational change as a function of temporal learning sequence. *47th Annual Conference of the Cognitive Science Society*.
- CA39 Khoudary A+, Peters MAK\*, **Bornstein AM\*** (2025). A principled model of uncertainty resolution explains dynamic effects of expectations in perceptual decision making. *Society for Philosophy and Neuroscience 2025*.  
**Selected for a talk.**

- CA38 Kapogianis T, **Bornstein AM**, Chrastil ER (2024). Size and community structure affect abstract graph learning. *45th Annual Conference of the Cognitive Science Society*.
- CA37 Azimihashemi M+, Banavar NV+, Guo S+, **Bornstein AM** (2024). Does alcohol dependence differentially influence episodic and semantic contributions to recognition priming in older adults? *SRNDNA Summer Symposium 2024*.
- CA36 Yamamoto A+, Guo S+, Nadkarni M+, Rouhani N, **Bornstein AM** (2024). Alcohol Prediction Errors in Individuals With Varying Levels of Alcohol Use. *SRNDNA Summer Symposium 2024*.
- CA35 Dinh ET+, Zhou D+, Guo S+, Noh SM+, Cooper KW, **Bornstein AM** (2024). Autoencoder models of human graph learning reveal that sparse and dense representations differentially support planning and recall *Society for Neuroscience Annual Meeting*.
- CA34 Harhen NC+, **Bornstein AM\***, Hartley CA\* (2024). Reinstated episodes and context differentially guide decision making across development. *Society for Neuroscience Annual Meeting*.
- CA33 Chen Y+, Harhen NC+, Stout DM, **Bornstein AM** (2024). Early-life unpredictability modulates planning horizon in a structured foraging task. *Society for Neuroscience Annual Meeting*.
- CA32 Yoo J+, **Bornstein AM** (2024). Goal-directed control evolves in tandem with multiple task representations. *Society for Neuroscience Annual Meeting*.
- CA31 Khoudary A+, **Bornstein AM\***, Peters MAK\* (2024). Reasoning goals and representational decisions in computational cognitive neuroscience: a case study from bounded evidence accumulation. *Philosophy and Neuroscience at the Gulf VII*.
- CA30 Martínez-Ortiz MA, Leonard B+, Yassa MA, **Bornstein AM**, Hartley CA (2024). Exploring the relationship between Early Life Adversity (ELA) dimensions and the development of decision-making strategies under motivational conflict. *The Flux Society Congress for Integrative Developmental Neuroscience*.
- CA29 Khoudary A+, Peters MAK\*, **Bornstein AM\*** (2024). Explaining dynamic effects of memory on perceptual decisions. *Learning and Memory 2024*.
- CA28 Khoudary A, **Bornstein AM\***, Peters MAK\* (2023). Characterizing dynamic effects of memory on perceptual evidence accumulation. *RIKEN CBS Summer Program*
- CA27 Chen Y+, Yoo J+, Harhen NC+, Noh SM+, Stout DM\*, **Bornstein AM\*** (2023). Adult and early-life adversity influences on memory-guided planning. *UCI Conte Center annual symposium*.
- CA26 Harhen NC+, Noh SM+, Stough-Lacking S+, **Bornstein AM** (2023). Suboptimal or locally rational? Foraging as a window onto the universal mechanisms of decisions under uncertainty. *Healing the Brain: Bridging the Gap in Low-and Middle-Income Countries*.
- CA25 Harhen NC+, **Bornstein AM** (2023). Temporal representation optimization provides a computational link between early life adversity and anhedonia. *Computational Psychiatry 2023*.
- CA24 Banavar NV+, **Bornstein AM** (2023). Decomposing behavioral pattern separation: A model-based analysis. *Learning and Memory 2023*.
- CA23 Harhen NC+, **Bornstein AM** (2023). Temporal representation adaptation as a computational link between early life unpredictability and anhedonia. *Learning and Memory 2023*.

- CA22 Khoudary A+, Peters MAK\*, **Bornstein AM\*** (2023). Characterizing dynamic effects of memory on perceptual decisions. *Learning and Memory 2023*.
- CA21 Noh SM+, **Bornstein AM** (2023). Memory encoding ability interacts with training interventions to improve memory-guided inference decisions *Learning and Memory 2023*.
- CA20 Yoo J+, **Bornstein AM** (2023). Humans build configural representations for planning in complex environments *Learning and Memory 2023*.
- CA19 Stout DM, Harhen NC+, **Bornstein AM**, Vinograd M, Spadoni A, Simmons AN, Yassa MA, Davis EP, Glynn LM, Baram TZ, Baker DG, Risbrough VB (2023). Unpredictable early-life experiences moderate the effect of anhedonia and PTSD symptoms on neural measures of reward learning in adulthood. *Anxiety and Depression Association of America*.
- CA18 Harhen NC+, **Bornstein AM** (2023). Interval timing as a computational framework for examining the pathway from early life unpredictability to affective disorders. *UCI Conte Center annual symposium*.  
**Best abstract award.**  
**Selected for a short talk.**
- CA17 Khoudary A+, **Bornstein AM\***, Peters MAK\* (2023). Perceptual decisions result from dynamic precision-weighted integration of memory and visual information. *Association of Scientific Studies of Consciousness*.  
**Selected for a talk.**
- CA16 Kapogianis T, **Bornstein AM**, Chrastil ER (2022). Graph Metrics and Non-Spatial Navigational Learning. *Society for Neuroscience Annual Meeting*.
- CA15 Harhen NC+, Hartley CA\*, **Bornstein AM\*** (2022). Memory-guided decision-making develops alongside model-based planning. *The Flux Society Congress for Integrative Developmental Neuroscience*.
- CA14 Harhen NC+, **Bornstein AM** (2022). Representation learning & adaptation in human foraging. *CNLM annual meeting*.  
**Selected for a short talk.**
- CA13 Yoo J+, **Bornstein AM** (2022). Task complexity and experience modulate the use of online planning. *CNLM annual meeting*.
- CA12 Noh SM+, Stark CEL, **Bornstein AM** (2022). Mnemonic Discrimination Ability Predicts Optimal Training Condition for Memory-Guided Inference Decisions. *Annual meeting of the Cognitive Neuroscience Society*.  
**Selected for a short talk.**
- CA11 Yoo J+, **Bornstein AM** (2021). Task complexity and experience dictate the use of online, versus offline, planning in humans. *Annual meeting of the Society for Neuroeconomics*.
- CA10 Noh SM+, Kerr T, Bennett IA\*, **Bornstein AM\*** (2021). Age-related differences in memory-guided decisions are driven by a trade-off between multiple decision systems. *Society for Neuroscience Annual Meeting*.
- CA9 Noh SM+, Kerr T, **Bornstein AM** (2021). Pattern Separation Predicts Which Memories Are Sampled During Decisions for Reward *Psychonomics*.
- CA8 Noh SM+, **Bornstein AM** (2021). Pattern separation mediates the types of memories sampled during decisions for reward. *CNLM annual meeting*.

- CA7 Banavar NV+, **Bornstein AM** (2021). Deliberative evaluation in intertemporal choice is shaped by experiment structure. *Annual meeting of the Society for Neuroeconomics*  
**Selected for a short talk.**
- CA6 Devine SM, Otto AR, **Bornstein AM\***, Louie K\* (2021). Context-dependent choice and evaluation in real-world consumer behavior *Mathematical Psychology*
- CA5 Harhen NC+, Yassa MA, Baram TZ, **Bornstein AM** (2021). Exploring a latent cause model of substance use disorder symptoms. *Biological Psychiatry*  
[doi:10.1016/j.biopsych.2021.02.479](https://doi.org/10.1016/j.biopsych.2021.02.479)
- CA4 Cooper KW+, Li L, Agostinelli F, Saraf M, Elias GA, Baldi P, **Bornstein AM**, Shahbaba B, Fortin N (2021). Theta-associated nonspatial sequence coding in hippocampus. *Society for Neuroscience Global Connectome*.
- CA3 Harhen NC+, Hartley CA, **Bornstein AM** (2020). Foraging behavior adjusts to multiple scales of context. *Annual meeting of the Society for Neuroeconomics*  
**Selected for a talk.**
- CA2 Vlasceanu M, Morais MJ, Zhao Z, **Bornstein AM**, Norman KA, Coman AC (2020). Self-Other Similarity Modulates the Socially-Triggered Context-Based Prediction Error Effect on Memory. *41st Annual Conference of the Cognitive Science Society*
- CA1 Rouhani N, Norman KA, Niv Y, **Bornstein AM** (2019). Reward prediction errors create event boundaries in memory *Psychonomics*

## AWARDS

|          |                                                                        |
|----------|------------------------------------------------------------------------|
| 2024     | CNLM Exceptional Mentor Award                                          |
| 2023     | Election to the Memory Disorders Research Society                      |
| 2020     | Association for Psychological Science “ <b>Rising Star</b> ” Award     |
| 2020     | Brain and Behavior Research Foundation NARSAD Young Investigator Award |
| 2005,6,8 | Honorable mention, NSF Graduate Research Fellowship                    |
| 2000     | Runner up, MIT \$50k Business Plan competition                         |

## TALKS

### *Invited*

|          |                                                                       |
|----------|-----------------------------------------------------------------------|
| Nov 2025 | Mechanistic Basis of Foraging, Birmingham, UK                         |
| Mar 2025 | Rutgers-Princeton Center for Computational Cognitive Neuro-Psychiatry |
| Feb 2025 | University of California, Berkeley                                    |
| Feb 2025 | National Institute of Drug Abuse                                      |
| Nov 2024 | Singidunum University, Belgrade, Serbia                               |
| Sep 2024 | New York University                                                   |
| Feb 2024 | University of California Los Angeles                                  |
| Feb 2024 | University of Miami                                                   |
| Jan 2024 | Virginia Tech                                                         |

|          |                                                               |
|----------|---------------------------------------------------------------|
| Dec 2023 | University of California Davis                                |
| Dec 2023 | Brown University                                              |
| May 2023 | Future of Foraging seminar series                             |
| Apr 2023 | Johns Hopkins University                                      |
| Mar 2023 | Northeastern University                                       |
| Feb 2023 | Center for the Neurobiology of Learning & Memory              |
| Dec 2022 | City University of New York                                   |
| Feb 2021 | University of California Los Angeles                          |
| Feb 2021 | Context and Affective Memory meeting                          |
| Aug 2020 | University of California Los Angeles                          |
| Mar 2020 | Claremont Colleges                                            |
| Feb 2020 | University of California Riverside                            |
| Sep 2019 | Facebook Research Labs                                        |
| May 2019 | University of California Los Angeles                          |
| Apr 2019 | Brown University                                              |
| Jan 2019 | National Institute of Drug Abuse                              |
| Dec 2017 | Cambridge University                                          |
| Oct 2017 | Johns Hopkins University                                      |
| Feb 2015 | Mount Sinai School of Medicine                                |
| Jun 2014 | Weill-Cornell Medical College                                 |
| Jun 2011 | Memory and Decisions meeting, Stanford University             |
| Jan 2011 | Parallel Distributed Processing meeting, Princeton University |
| Jan 2011 | Kavli Institute, Harvard University                           |

*Contributed*

|          |                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Apr 2025 | Cognitive Neuroscience Society symposium, “Uncertainty Resolution across Learning, Memory, and Decision-making”                               |
| Jan 2025 | Workshop on Self-assembling Games, UC Irvine                                                                                                  |
| May 2024 | American Psychological Society (Presenter: +Sharon M Noh)                                                                                     |
| May 2022 | Center for the Neurobiology of Learning and Memory Annual Meeting                                                                             |
| Apr 2021 | Society of Biological Psychiatry symposium “Advancing Treatment of Substance Use Disorders Through Computational Modeling”                    |
| Jan 2020 | Winter Conference on the Neurobiology of Learning and Memory symposium “Mnemonic contributions to reinforcement learning and decision making” |
| Mar 2018 | Cosyne workshop “Hippocampal computations and interactions supporting statistical learning and decision-making”                               |
| Nov 2015 | SFN symposium “Understanding Goal-Directed Decision-Making in Humans: Computations and Circuits”                                              |
| Mar 2014 | Workshop on the Neurobiology of Prediction and Surprise, Rutgers University                                                                   |

TEACHING/SERVICE

## *UC Irvine*

|                              |                                                                        |
|------------------------------|------------------------------------------------------------------------|
| Spring 2027                  | Neural Data Analysis (Undergraduate)                                   |
| Fall 2026                    | Cognitive Maps (Graduate; cross-listed with Critical Theory Emphasis)  |
| Fall 2025                    | Resisting “Neuro-narratives” (Undergraduate)                           |
| Winter 2025,2027             | LIFTED UC Irvine Prison Education Program (Human Memory)               |
| Spring 2023-2026,Winter 2027 | Human Memory (Undergraduate)                                           |
| Spring 2023-24,Fall 2025     | Decision making (Graduate)                                             |
| Spring 2021-22               | Decision making & Problem solving (Graduate; with Prof. Zygmunt Pizlo) |
| Spring 2020                  | Topics in Reinforcement Learning (Graduate; with Prof. Mimi Liljeholm) |
| Spring 2019-22               | Research in Exp Psych (Undergraduate; with Prof. Nadia Chernyak)       |
| Winter 2019-22               | Advanced Experimental Psychology (Undergraduate)                       |

## *Other teaching/mentorship*

|                |                                                                     |
|----------------|---------------------------------------------------------------------|
| Fall 2020      | Neuromatch academy mentor                                           |
| Summer 2020    | Neuromatch academy group leader                                     |
| Summer 2020    | GSMI Cientifico Latino                                              |
| 2019           | MEET alumni mentor                                                  |
| Summer 2018    | Computational & Cognitive Neuroscience Summer School, Suzhou, China |
| 2016-2018      | Princeton Prison Teaching Initiative                                |
| Summer 2007,08 | MIT Middle East Education through Technology (MEET), Jerusalem      |

## *University/Department service*

|               |                                                                                        |
|---------------|----------------------------------------------------------------------------------------|
| 2024-         | UCI Research Imaging Equipment Committee                                               |
| 2022-2025     | Cognitive Sciences Colloquium committee, faculty advisor                               |
| 2021-         | Irvine Faculty Association, Executive board member; 2023-2025 Treasurer; 2025-Co-Chair |
| 2020          | CNLM “Evening to Remember” organizational committee                                    |
| 2019-         | First-Generation Faculty Initiative                                                    |
| 2019-         | UCI Prison Education Program, advisory board, curriculum committee                     |
| 2019,23       | Cognitive Sciences faculty search committee                                            |
| 2018,20,22,24 | Cognitive Sciences PhD admissions committee                                            |

## *Organizational/Field service*

|       |                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------|
| 2025- | Associate Editor, <i>Cognitive Science</i> .                                                    |
| 2025- | Justice, Diversity, Equity, and Inclusion Committee, <i>Memory Disorders Research Society</i> . |
| 2025- | Advisor, <i>All People’s Health Collective</i> .                                                |
| 2024- | Editorial board, <i>Translational Neuroscience</i> .                                            |
| 2023  | UCI Conte Center annual symposium, organized with the Conte Center Team.                        |

2022     NSF/Simons NeuroDataScience workshop, Co-Organizer (with Norbert Fortin and Babak Shahbaba)

2021     Center for Neurobiology of Learning and Memory Spring Meeting Co-organizer (with Lulu Chen)

2018     Princeton Neuroscience Institute “Inside-Out” seminar series Co-organizer (with Ahmed El Hady)

2018     “Goal-Directed Decision Making: Computations and Circuits” *Elsevier* Co-editor (with Richard Morris & Amitai Shenhav)

2015     COSYNE Workshop “Memory in action: The role(s) of the hippocampus in decisions for reward” Co-organizer (with G. Elliott Wimmer)